rORForise: Base-by-base evaluation of gene predictions on unassembled reads

Nicholas J. Dimonaco [1,2], Wayne Aubrey [2], Michael G. Surette [3,4], Amanda Clare [2]
[1]Institute for Global Food Security, School of Biological Sciences,
Queen's University Belfast, Belfast, UK
[2]Department of Computer Science, Aberystwyth University, Aberystwyth, UK
[3]Farncombe Family Digestive Health Research Institute, McMaster University, Canada
[4]Department of Medicine, McMaster University, Canada

Abstract

Gene prediction is a critical step in converting sequencing data into a functional description of a genome or metagenome. Evaluation of gene predictors, however, has traditionally been performed on assembled genomes or other contiguous sequences, where predicted coding sequences can be compared directly with annotated reference genes. Short-read metagenomes create a different evaluation problem: predictions are made on fragmented, unassembled sequences, and an individual prediction may represent only part of a true coding sequence.

Here we introduce *rORForise*, a lightweight framework for evaluating coding-sequence predictions made directly from unassembled reads. Inspired by the reference-based, multi-metric evaluation philosophy of ORForise, rORForise moves the ground-truth comparison to the read level. Reads are mapped to a reference genome with a trusted CDS annotation, allowing the genomic position represented by each read to be established. Read-level CDS predictions can then be evaluated base-by-base against the reference annotation, distinguishing sequence that is supported by a true coding region from sequence predicted outside the annotated coding space. The framework also retains feature-level measures such as detection, start and stop accuracy, partial predictions and unsupported prediction sequence.

The current release focuses on the evaluation framework and supporting utilities rather than a completed benchmark of prediction tools. Systematic benchmarking across genomes, read lengths, sequencing-error profiles and prediction methods is ongoing. rORForise provides a simple, transparent basis for quantifying where read-level gene predictions are supported, incomplete or incorrect, and for comparing prediction methods using the same underlying sequencing evidence.

Introduction

Accurate coding-sequence (CDS) prediction is a prerequisite for many downstream genomic analyses, including functional annotation, comparative genomics and estimation of gene content. The ORForise framework demonstrated the value of evaluating prokaryotic gene prediction tools against a common genome-level reference using multiple complementary metrics rather than reducing performance to a single accuracy value [1]. Its central principle is simple: predictions should be evaluated against an independently defined biological reference using metrics that expose different forms of disagreement.

The same principle is harder to apply to unassembled sequencing reads. A short read may cover only a small part of a coding sequence, may begin or end within a CDS, or may contain sequence immediately outside the CDS. Read-level gene prediction tools therefore do not return the same kind of object as genome-level predictors: a prediction may be a partial coding fragment rather than a complete gene model. Evaluation based only on whether the predicted feature exactly matches a reference CDS is consequently insufficient.

This problem is increasingly relevant to metagenomics, where large numbers of reads may remain unassembled or may not be represented in a final assembly. Rather than treating such reads as inherently unevaluable, we sought to adapt the reference-based philosophy of ORForise to the read level. The key observation is that an assembled reference genome can provide a positional ground truth even when the prediction itself is made without an assembly. If sequencing reads are mapped to the reference genome, each base in each read

can be assigned a genomic coordinate and therefore classified according to whether it falls within an annotated CDS.

rORForise implements this principle. It is designed as a predictor-agnostic evaluation framework for read-level coding-sequence predictions. The primary objective is not to call genes, recover genomes or determine whether a reference annotation is biologically complete. Instead, it provides a reproducible way to ask a narrower question: *given unassembled reads originating from a genome with an independent CDS annotation, how much of a predicted coding sequence is supported by the underlying reference coding regions?*

Materials and Methods

Evaluation principle

rORForise separates gene prediction from gene prediction evaluation. A predictor is first run on unassembled reads exactly as it would be used in a read-level analysis. Independently, the same reads are mapped to a reference genome for which a CDS annotation is available. The reference annotation is treated as the evaluation ground truth for genomic coding coordinates.

The evaluation can therefore be represented as:

1. a reference genome and trusted CDS annotation define the ground-truth coding regions;
2. unassembled reads are generated or obtained independently;
3. a read-level CDS predictor generates candidate coding features on those reads;
4. the reads are mapped back to the reference genome;
5. the genomic coordinates represented by each predicted feature are established from the read-to-genome alignments;
6. predicted bases are compared base-by-base with the reference CDS coordinates; and
7. base-level comparisons are aggregated into feature- and dataset-level measures.

The important distinction is that the prediction remains read-based throughout. The reference genome is used only to establish where the sequence came from and whether each predicted nucleotide is consistent with an annotated CDS.

Reference genomes and annotations

A reference genome is required for the ground-truth evaluation mode. The reference consists of a genome FASTA file and an independently produced GFF annotation containing CDS features. In development, the framework has been tested with bacterial genomes including *Escherichia coli*, *Staphylococcus aureus* and *Mycoplasma genitalium*. The latter provides a useful stress case because its genetic code differs from the standard bacterial code, allowing codon-table handling to be considered separately from the evaluation framework itself.

The framework does not require a particular annotation provider. Ensembl Bacteria or another curated genome annotation can be used as the reference provided that the CDS coordinates are available in a consistent format. The reference is treated as a fixed evaluation target rather than as a prediction generated by rORForise.

Read generation and mapping

For controlled experiments, the package includes *Fragment-Genome*, a utility for generating read-length fragments from a reference genome while retaining the genomic coordinates represented by each fragment. Synthetic reads can additionally be generated with established read simulators such as ART [5]. This provides an exact correspondence between the simulated read sequence and its source genome coordinates.

For observed sequencing data, reads are mapped directly to the reference genome using a standard aligner such as Bowtie2 [6]. The resulting alignments provide the coordinate transformation required for evaluation. The reference genome and its CDS annotation therefore remain external to the read-level predictor and are introduced only at the evaluation stage.

Read-level CDS prediction

rORForise is predictor-agnostic. The current development workflow has been used with read-level prediction approaches including FragGeneScan [2], Pyrodigal [4] and simple ORF enumeration approaches. The framework does not depend on a particular prediction algorithm or on a particular output format beyond the ability to represent predicted CDS coordinates on the input reads.

This separation is intentional. rORForise evaluates the prediction produced by a tool rather than

changing the prediction process itself. Different predictors can therefore be run on the same read set and evaluated against the same reference evidence.

Read–CDS intersection and coordinate projection

The rORForise package includes *bam-gff-intersect*, which intersects mapped reads with reference CDS features. For each aligned read, the procedure identifies the positions at which the read overlaps an annotated coding region. These intersections provide the genomic evidence needed to determine whether sequence predicted as coding on the read lies within a reference CDS or outside annotated coding sequence.

For simulated data, the known origin of each fragment can be retained directly. For mapped experimental reads, genomic coordinates are obtained from the read-to-reference alignment. The same evaluation framework can therefore operate with either simulated or observed reads.

Base-by-base evaluation

The central output of rORForise is a nucleotide-level comparison between predicted coding sequence and reference coding sequence. For every nucleotide position represented by a predicted feature, the framework asks whether the corresponding genomic position is annotated as CDS in the reference.

This produces a sequence of per-base classifications that can be aggregated into measures including:

- the fraction of predicted bases supported by reference CDS sequence;
- the fraction of reference CDS bases detected by predictions;
- unsupported predicted sequence outside annotated CDS regions;
- missing coding sequence within partially predicted CDSs; and
- the positions of disagreements relative to CDS starts and stops.

These quantities allow partial predictions to be evaluated without requiring them to match a complete reference CDS. They also allow errors at the beginning and end of predicted features to be separated from errors within the coding region itself.

The resulting base-level measurements can be summarised at several levels. At the feature level, predictions can be classified as supported, partially supported or unsupported according to predefined overlap criteria. At the dataset level, the framework can report aggregate precision, recall and coverage-style measures together with the distribution of start and stop discrepancies. The precise thresholds used for feature-level summaries remain configurable so that alternative definitions of a successful prediction can be tested without changing the underlying nucleotide-level evidence.

Relationship to genome-level evaluation

ORForise evaluates gene predictions on contiguous genomic sequences and provides a multi-dimensional framework for comparing predicted gene models with an annotated reference [1]. rORForise adopts this evaluation philosophy but changes the unit of comparison. Instead of comparing complete genome-level CDS models directly, it evaluates the sequence represented by individual reads and projects the prediction onto the reference genome through the read alignment.

This distinction is important because read-level predictions are inherently partial. A predicted feature can be correct over the sequence it covers while representing only part of a reference CDS. Conversely, a predictor can produce a plausible-looking ORF fragment that extends outside the true CDS boundaries. A base-level representation preserves these distinctions rather than reducing both cases to a binary correct/incorrect label.

Evaluation plan

The current rORForise release is intended primarily as a methods preprint. Systematic benchmarking is being developed using multiple bacterial genomes, read lengths and sequencing-error profiles and comparing alternative read-level CDS predictors on identical read sets. Planned analyses will examine nucleotide-level support, CDS detection, start and stop accuracy, partial predictions, unsupported predicted sequence and the effect of read length and sequencing error.

A key aim is to make these measurements directly comparable across prediction methods. Because all predictors are evaluated against the same underlying mapped reads and the same reference CDS coordinates, differences can be attributed to the behaviour of the prediction methods rather than

to differences in the input sequence or evaluation procedure.

No claims are made here that one read-level predictor is currently superior to another. The purpose of the present work is to establish the evaluation framework needed to measure those differences systematically.

Scope and limitations

rORForise evaluates coding-sequence prediction; it does not establish that the reference annotation is complete or biologically correct. Its ground truth is therefore only as good as the reference genome and CDS annotation used for evaluation. This is appropriate for controlled method benchmarking, but limits direct inference about genes absent from the reference annotation.

The framework also evaluates the sequence that can be supported by the read and its genomic alignment. Where a read does not contain sufficient sequence to distinguish alternative gene models, rORForise does not infer the missing sequence. Likewise, repetitive or ambiguously mapped reads may require explicit filtering or confidence thresholds. These limitations are part of the evaluation problem and can be reported alongside prediction performance.

Software availability

rORForise is available at `https://github.com/NickJD/rORForise`. The package contains the read-processing and intersection utilities used by the evaluation workflow, including *FragmentGenome* and *bam-gff-intersect*. The software is intended to provide a transparent, reproducible basis for evaluating read-level CDS prediction methods.

Conclusion

rORForise adapts the reference-based, multi-metric philosophy of ORForise to the problem of CDS prediction on unassembled reads. By mapping reads to an annotated reference genome and evaluating predicted coding sequence base-by-base against the reference CDS coordinates, the framework preserves information that is lost when read-level predictions are judged only by complete-feature matching. The immediate purpose of the software is methodological: to provide a simple and reproducible way to measure where read-level gene predictions are supported, partial or unsupported. Systematic benchmarking using this framework will establish the strengths and limits of different approaches to coding-sequence prediction on fragmented sequencing data.

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Conflict of interest statement.

None declared.

References

[1] Dimonaco NJ, Aubrey W, Kenobi K, Clare A, Creevey CJ. No one tool to rule them all: prokaryotic gene prediction tool annotations are highly dependent on the organism of study. *Bioinformatics*. 2022;38(5):1198–1207. doi:10.1093/bioinformatics/btab827.

[2] Rho M, Tang H, Ye Y. FragGeneScan: predicting genes in short and error-prone reads. *Nucleic Acids Res*. 2010;38(20):e191. doi:10.1093/nar/gkq747.

[3] Hyatt D, Chen G-L, Locascio PF, Land ML, Larimer FW, Hauser LJ. Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC Bioinformatics*. 2010;11:119. doi:10.1186/1471-2105-11-119.

[4] Larralde M. Pyrodigal: Python bindings and interface to Prodigal, an efficient method for gene prediction in prokaryotes. *J Open Source Softw*. 2022;7(72):4296. doi:10.21105/joss.04296.

[5] Huang W, Li L, Myers JR, Marth GT. ART: a next-generation sequencing read simulator. *Bioinformatics*. 2012;28(4):593–594. doi:10.1093/bioinformatics/btr708.

[6] Langmead B, Salzberg SL. Fast gapped-read alignment with Bowtie 2. *Nat Methods*. 2012;9(4):357–359. doi:10.1038/nmeth.1923.

[7] Dimonaco NJ, Clare A, Aubrey W, Surette MG. rORForise: read-based gene coverage evaluation. Software repository. 2026. Available at: `https://github.com/NickJD/rORForise`.